Ratios & Proportions Review

Sections 7.1 & 7.2

Name: _____

Date: _____ **Period:** _____

1 WARM-UP:

1. **Circle True or False for each statement.**

Statement	T / F
In the proportion $\frac{a}{b} = \frac{c}{d}$, the values b and c are called the **extremes**.	T F
$\frac{a}{b} = \frac{c}{d}$ and $\frac{b}{a} = \frac{d}{c}$ are equivalent proportions.	T F
An extended ratio can compare more than two quantities.	T F
If cross products are *not* equal, the proportion is true.	T F

2. **Circle the largest ratio (show your work/reasoning)**:

$$\frac{3}{8} \qquad \frac{5}{6} \qquad \frac{7}{12}$$

2 LESSON REVIEW:

Complete the definitions below as we review them together.

7.1 Vocabulary

1. **Ratio:** A _____ of two quantities.

   Can be written three ways: _____ or _____ or "a to b."

   Important rule about units: _____

   _____

2. **Proportion:** An _____ stating that two _____ are _____.

   Written as: _____

3. **Extended ratio:** Compares _____ quantities.

   Example: _____

4. **In a *True* proportion:** Cross products are *Equal* / *Unequal* (circle one)

   In a *False* proportion: Cross products are *Equal* / *Unequal* (circle one)

5. Consider the proportion $\frac{a}{b} = \frac{c}{d}$.

   Means: _____ and _____

   Extremes: _____ and _____

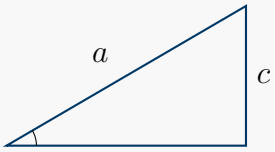
   Cross products: _____ and _____

7.2 Vocabulary: Equivalent Proportions

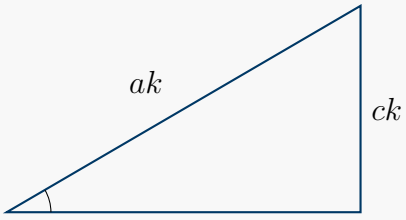
Starting from $\frac{a}{b} = \frac{c}{d}$, complete each equivalent proportion:

Operation	Equivalent Proportion	True or False?
Switch the means	$\frac{a}{\quad} = \frac{\quad}{d}$	
Take reciprocals of each ratio	$\frac{\quad}{a} = \frac{\quad}{c}$	
Add denominator to each numerator	$\frac{a \ +}{b} = \frac{c \ +}{d}$	

Proportions in Triangles: When corresponding sides of two triangles are in proportion, each pair of corresponding sides shares the *same ratio*. Scaling every side by the same factor k preserves all internal ratios.



$$\frac{a}{b} = \frac{7}{12}$$



$$\frac{ak}{bk} = \frac{a}{b} = \frac{7}{12}$$

Key idea: If $a = 7$, $b = 12$, and $k = 5$, then compute:

$$\frac{b}{a} = _____ \quad \text{and} \quad \frac{ak}{bk} = _____$$

3 PRACTICE PROBLEMS: *(8 minutes)*

Work independently and show your work. Then we'll share at the whiteboard.

1. Is $\frac{15}{25} = \frac{12}{20}$ a true or false proportion? *Justify your answer using cross products.*

Answer: True / False (circle one)

2. Solve for x in the given proportion:

$$\frac{x+2}{5} = \frac{x-1}{3}$$

Answer: _____

3. On a road map, **1 inch** represents **25 miles**. Two cities are $3\frac{1}{2}$ **inches** apart on the map.
   (a) Draw a picture of the map.
   (b) Solve a proportion to find the actual distance x (in miles) between the two cities.

Answer: _____ miles

Two Different Roles for Units in Proportions

Case 1: Two different kinds — units stay as-is

Each ratio pairs two different *kinds of quantity linked by a fixed scale (e.g. map distance and real distance).*

$$\frac{1\,\text{in}_{\text{map}}}{25\,\text{mi}} = \frac{3.5\,\text{in}_{\text{map}}}{d\,\text{mi}}$$

$\downarrow$ *no conversion needed*

$$d = 87.5\,\text{mi}$$

The proportion works because both ratios express the *same scale factor*. "Inches" and "miles" are *supposed* to be different — that is the scale.

Case 2: Same kind of quantity — units must match

Both quantities measure the same thing (e.g. both are lengths). Before writing the ratio, convert to a common unit.

$$3\,\text{ft} : 9\,\text{in}$$

$\downarrow$ *convert first*

$$36\,\text{in} : 9\,\text{in} = 4$$

Comparing 3 and 9 directly ignores that $1\,\text{ft} = 12\,\text{in}$. The ratio is only meaningful once both sides share a unit.

Rule of thumb:
Two different kinds, fixed scale? $\rightarrow$ set up the proportion directly.
Same kind of quantity? $\rightarrow$ convert first.

4 THE GOLDEN RATIO: Introduction & Group Activity

4.1 Where does φ come from?

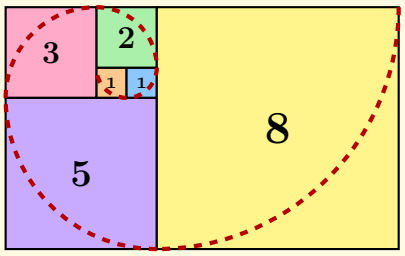
The Fibonacci Numbers are a sequence of whole numbers where each number is the sum of the two before it (known as a recursive sequence):

$$1, \quad 1, \quad 2, \quad 3, \quad 5, \quad 8, \quad 13, \quad 21, \quad 34, \quad 55, \quad 89, \quad 144, \quad \ldots$$

These numbers appear throughout nature — in the branching of trees, the arrangement of seeds in a sunflower, and the spiral of a nautilus shell and of spiral galaxies.

Now look at the **ratios of consecutive Fibonacci numbers**:

$$\frac{2}{1} = 2.000 \qquad \frac{13}{8} = 1.625$$
$$\frac{3}{2} = 1.500 \qquad \frac{21}{13} \approx 1.615$$
$$\frac{5}{3} \approx 1.667 \qquad \frac{34}{21} \approx 1.619$$
$$\frac{8}{5} = 1.600 \qquad \frac{55}{34} \approx 1.618$$



The Fibonacci spiral, built from squares whose side lengths are Fibonacci numbers, approximates the ***Golden Spiral*** *found in nature.*

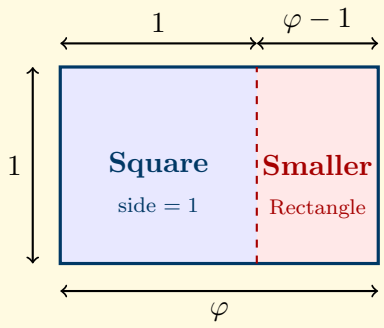
The ratios **converge** to a single value, the **Golden Ratio**:

$$\varphi \approx 1.618\ldots$$

4.2 The Golden Rectangle

The Golden Ratio also defines the **Golden Rectangle**: a rectangle whose long side and short side are in the ratio $\varphi : 1$. Its defining property is this — if you cut off the square from one end, the *remaining small rectangle* has exactly the *same proportions* as the original. That is:

$$\frac{\text{long side}}{\text{short side}} = \frac{\text{short side}}{\text{long side} - \text{short side}} \quad \Longrightarrow \quad \frac{L}{S} = \frac{S}{L - S}$$



Thus, for the Golden Rectangle, we have:

$$\frac{\varphi}{1} = \frac{1}{\varphi - 1}$$

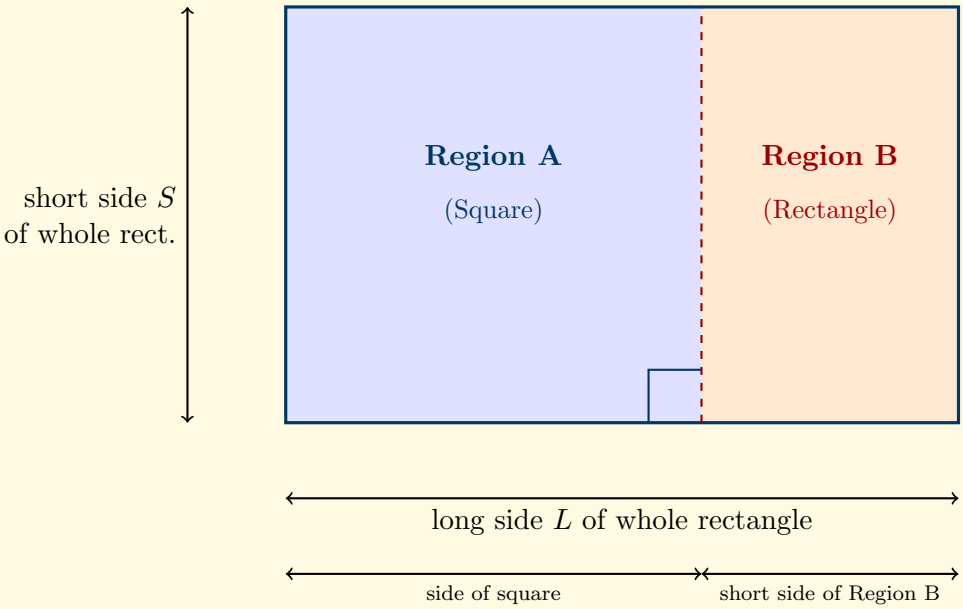
Mathematically verify the above proportion is true by substituting the value of $\varphi = 1.618$ and simplifying the ratios.

On the next page, you will **measure a real Golden Rectangle** printed on the page and **physically verify** this proportion using cross products and equivalent proportions.

4.3 Group Activity — *work in groups of 3 or 4*

Use a ruler and the shapes printed below. **All measurements are in millimeters.**

There are 3 shapes: the whole rectangle, Region A (square), and Region B (small rectangle).



1. **Measure** the whole rectangle with your ruler and record:

   Long side $L =$ _____ mm Short side $S =$ _____ mm

2. **Write the ratio** of long side to short side and evaluate (round to 3 places):

   $$\frac{L}{S} = \text{_____} \approx \text{_____}$$

3. **Measure Region B** (the smaller rectangle to the right of the dashed line):

   Long side of B, = _____ mm Short side of B = _____ mm

4. Write the ratio of Region B's **long side to short side** and evaluate:

   $$\text{_____} \approx \text{_____}$$

5. **Set up a proportion** stating that the ratio of the whole rectangle (Step 2) equals the ratio of Region B (Step 5). Verify it is approximately a **true proportion** using cross products. Show all work.

   Answer: True / False (circle one)

6. **Apply one equivalent proportion rule** from Section 7.2 to your proportion in Step 6. Write the new proportion and verify it is also true.

   Rule used: _____

Group members: _____

5 BRIDGE TO SIMILAR TRIANGLES

A Puzzle. Your teacher will draw three triangles on the board — each a different size, but all with the *same* angles. Record the side lengths in the table below.

Triangle	**Side** a	**Side** b	**Side** c	$\frac{a}{b} =$
Small				
Medium				
Large				

1. What do you notice about the ratio $\frac{a}{b}$ across all three triangles?

2. Is $\frac{a}{c}$ the same across all three triangles? Check it and record your finding below.

3. **Why** do you think the ratio stays the same when the triangles have the same angles but different sizes? Think about what "scaling" a triangle means.

4. **Prediction:** If a fourth triangle had the same angles but side $a = 15$, and you know $\frac{a}{b} = \frac{2}{3}$ from the table, find side b.

   $b =$ _____

5. What would happen to all the ratios if you *changed* one of the angles to be differnt between the triangles? Make a guess — you will find out in the next unit.

   _____

   _____

Big Idea: Two triangles that share the same angles *always* have the same side ratios — no matter how large or small they are. The next unit will give this powerful idea a name!